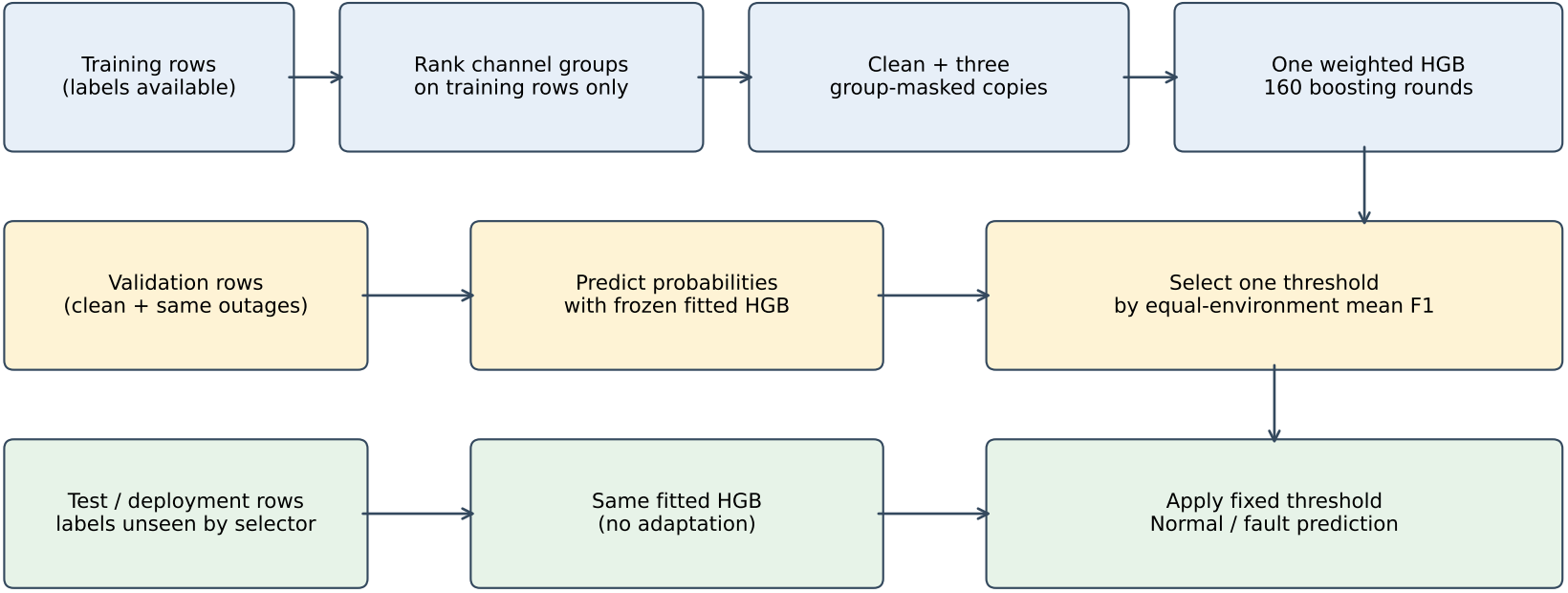
Training rows
(labels available)
Rank channel groups
on training rows only
Clean + three
group-masked copies
One weighted HGB
160 boosting rounds
Validation rows
(clean + same outages)
Predict probabilities
with frozen fitted HGB
Select one threshold
by equal-environment mean F1
Test / deployment rows
labels unseen by selector
Same fitted HGB
(no adaptation)
Apply fixed threshold
Normal / fault prediction